suggests that, to be a successful long-run investor, you should start off being a successful short-run investor. After doing this, take on all the advantages that the long horizon gives you.

I now discuss one important case where there is no difference between long-run investors and short-run investors. This case happens to be the most empirically relevant and is the foundation of any long-term investment strategy.

2.4. REBALANCING

Suppose that returns are not predictable, or the investment opportunity set is independent and identically distributed (i.i.d.). The i.i.d. assumption is very realistic. Asset returns are hard to predict, as chapter 8 will show. A good way to think about the i.i.d. assumption is that asset returns are like a series of coin flips, except coins can only land heads or tails, and returns can take on many different values. The coin flip is i.i.d. because the current probability of a head or tail does not depend on the series of head or tails realized in the past. The same is true for asset returns: when asset returns are i.i.d., in every period returns are drawn from the same distribution that is independent of returns drawn in previous periods. Under i.i.d. returns, assets are glorified coin flips.

With i.i.d. returns and a fixed risk-free rate, the dynamic portfolio problem becomes a series of identical one-period problems, as shown in Figure 4.4. If returns are not predictable, then the long-horizon portfolio weight is identical to the myopic portfolio weight. Put another way, with i.i.d. returns, there is no difference between long-horizon investing and short-horizon investing: all investors are short term, and it does not matter what the horizon is. We can write this as:

$$\text{Long-Run Weight } (t) = \text{Short-Run Weight } (t). \tag{4.13}$$

The short-run weight is the myopic portfolio weight in equation (4.5). It is stated in terms of CRRA utility, but more generally it is the portfolio weight of

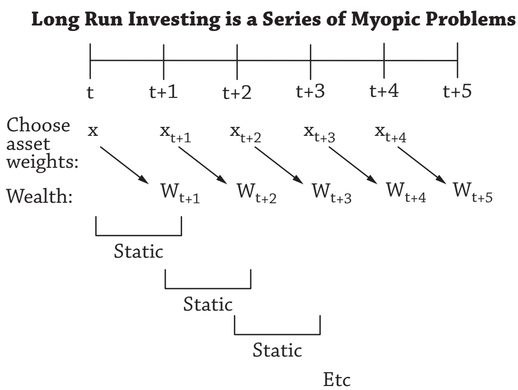

Figure 4.4